

① Introduction to HTML

Page No.

Date. / /

What is HTML?

- HTML (Hyper Text Markup Language) is the standard language used to create and structure content on the web.
- HTML tells web browsers how to display text, images, links, and other elements on a web page.

What does HTML do?

HTML uses tags (also called elements) to structure content (tags root of html)

ex:-

- ① Add headings and paragraphs
- ② Insert images or videos
- ③ Create links and buttons
- ④ Make lists and tables
- ⑤ Organize layout using sections or divisions (divs)

ex:-

```
<html>
```

```
<head>
```

```
<title>
```

```
</title>
```

```
</head>
```

```
<body>
```

```
<p> this is college </p>
```

```
<h1> this is web tech </h1>
```

```
<h2> This is web tech </h2>
```

```
</body>
```

```
</html>
```


History

- HTML is the standard markup language used to create web pages
- HTML was invented by Tim Berners-Lee in 1991

Evolution of HTML Versions

HTML 1.0 → HTML 2.0 was published in 1995

HTML 3.0 → HTML 3.0 was introduced by Dave Raggett in 1995

HTML 4.01 → released in late 1999

HTML 5 → was published as a W3C Recommendation in 2014. It introduced new elements and attributes, such as `<canvas>`, `<video>` and `<audio>`

What is an HTML Document?

- An HTML document is a text file that contains HTML code
- It tells a web browser how to display a web page
- These files usually have a .html or .htm extension

Lab Program -1

Basic structure of an HTML Document

```
<!DOCTYPE html>
<html>
<head>
<title> My first HTML Page </title>
</head>
<body>
<h1> Welcome to my website </h1>
<p> This is my first paragraph in HTML </p>
</body>
</html>
```

Explanation

- `<!DOCTYPE html>` - used to declare document type
- `<html>` - Tells the browser that this is an HTML document
- `<head>` - contains metadata (ex: title, style, scripts)
- `<title>` - sets the title shown on the browser tab
- `<body>` - contains the content that appears on the page

src attribute

used to link external resources like images, CSS, JavaScript, etc.

Example: ``

only used in opening tag.

versions of HTML

① HTML 1.0 - 1993

HTML has evolved significantly over the years from its initial release as HTML 1.0 to its latest version HTML 5.

② HTML 1.0 - 1993

- It was the first standard version of HTML.
- Limited functionality, focused on structuring and displaying text documents.
- Basic elements like headings, paragraphs, lists, and links were introduced.

③ HTML 2.0 - 1995

- Introduced to support for tables, allowing better organisation of data.
- The concept of frames was introduced, enabling user input and interactions with web pages.

④ HTML 3.2 - 1997

- Enhanced support for tables, forms and text formatting.
- Introduction of new elements like image maps, and basic support for style sheets.

⑤ HTML 4.01 - 1999

- Improve support for multimedia elements like audio and video.
- Introduction of frames, allowing the division of web pages into multiple sections.

Expanded support for style sheets and scripting through CSS and JavaScript.

no support required to play audio / video.

⑥ HTML 5 - 2014

- Significant advancements over its predecessors bringing major changes and new features.
- Introduced a more symmetric approach to markup with elements like `<header>`, `<footer>`.
- Support for multimedia elements without requiring plugins (e.g., `<audio>`, `<video>` tags).
- Improved form input types including email, URL, date and range.
- Introduced the `<canvas>` element/tag for dynamic, scriptable rendering of 2D shapes and bitmap images.

Element - Tags

- HTML elements are represented by tags.
- The opening tag starts with the element name, followed by any attributes within the tag.
- The closing tag is the same as the opening tag, but with a forward slash (!) before the element name.

Example: `<elementName attribute="value">... </elementName>`

Attribute

- Attributes provide additional information or modify the behaviour of HTML elements.
- They are placed within the opening tag of an element.
- Attributes consist of a name and a value separated by an equal sign (=) and enclosed in quotes.

Example: `<elementName attribute="value">`

`</elementName>`

Comments

- HTML comments are used to add notes or explanations to the code and are not displayed in web pages.
- They are enclosed within `<!--` and `-->`.

ex: `<!-- This is a comment -->`

Self Closing Tags (Empty tags)

- Some elements (tags) such as line breaks, or images do not require a closing tag.
- They can be self closed by using a forward slash before closing angle bracket.

ex: `<br />` → break tag.

`<img src = "image.jpg" alt = "Image" />`

Creating an HTML Document

Steps to create an HTML Document

- Steps: Open a text editor
- Use any text editor like: Notepad, VS code, Sublime Text, Atom (for more features)

Steps: Write Basic HTML code

`<!DOCTYPE html>`

`<html>`

`<head>`

`<title> My first HTML page </title>`

`</head>`

`<body>`

`<h1> Welcome to my website </h1>`

`<p> This is my first paragraph in HTML </p>`

`</body>`

`</html>`

make-up language → HTML

HTML

Step 3: Save the file

- Go to File > Save as
- Save the file with the name: index.html
- Choose UTF-8 encoding (important for special characters)

Step 4: Open in a browser

- Find your saved file
- Double click it
- Your default browser will open and display the content and it will not be a 404 error

HEADINGS AND PARAGRAPHS IN HTML

1. HEADINGS in HTML

HTML provides six levels of headings from `<h1>` (largest/smallest) to `<h6>`

ex: `<h1>` This is a level 1 heading `</h1>`

`<h2>` This is a level 2 heading `</h2>`

`<h3>` This is a level 3 heading `</h3>`

`<h4>` This is a level 4 heading `</h4>`

`<h5>` This is a level 5 heading `</h5>`

`<h6>` This is a level 6 heading `</h6>`

2. PARAGRAPHS in HTML

The `<p>` tag is used to define a paragraph of text

ex: `<p>` This is my first paragraph. `</p>`

Line Breaks in HTML

- In HTML, to force text to move to the next line without starting a new paragraph use the `<br>` tag
- `<br>` stands for "break"
- It is an empty (self-closing) tag - it doesn't need a closing `</br>`
- Usually inside paragraph or when formatting poems, address or forms

Example:- `<!DOCTYPE html>`

`<html>`

`<head>`

`<title> Line Break Example </title>`

`<body>`

`<a href="#"> Add URL </a>`

`<p> Arjun <br> 123 Main Street <br> Kollam </p>`

`</body>`

`</html>`

Arjun

123 Main Street

Kollam

Chapter - 2

2. Elements of HTML

- Elements (Tags) of HTML -> will define the structure and content of a web page
- Every part of a webpage (like text, images, links, or sections) is made up of elements

WORKING WITH TEXT IN HTML

- When building a webpage one of the first things we do is work with text - adding titles, paragraphs, line breaks, emphasis and more
- HTML provides several elements to format and structure text

Common Text Elements in HTML

(13)

Tag	Description	Example
<code><h1></code> to <code><h6></code>	Headings (h1 is largest, h6 is smallest)	<code><h1> Main Title </h1></code>
<code><p></code>	Paragraph	<code><p> This is a paragraph </p></code>
<code>
</code>	Line Break	<code>
 This is a line
 break</code>
<code></code> / <code></code>	used to make text bold	<code> Important text </code> or <code></code>
<code></code> / <code><i></code>	Used to make text italic	<code> Emphasized text </code> or <code><i></code>

Tag	Description	Example
⑥ <u> / </u>	used to underline text	<p> this text is <u> underlined </u>
⑦ <s>	used to add a strike through elment to text	<p> this text is <s> strikethrough </s>
⑧ <sub>	used to create subscript text	H₂O
⑨ <sup>	used to create superscript text	π²
⑩ <mark>	Highlight Text	<mark> Highlighted </mark>
⑪ <small>	Decrease the size of the text by one level	footnote
⑫ <big>	Increase the size of the text by one level	<big> header </big>
⑬ 	Text that has been deleted from a document becomes with usually strike a line through deleted text	<p> My favourite color is blue

II Working With Lists in HTML

Lab Program-5

Lists are used in HTML to display multiple items in structured way - such as menu, steps, to do items or collections

Two main types of lists in HTML:

- ① Unordered list -
- ② Ordered list -
- ③ Description list

1. UNORDERED LIST ()

- An unordered list is created using the '' tag and each list item is defined with '' tag
- The list items will be marked with bullet (small black circles) by default
- Use when order doesn't matter

Ex -
 Apple
 Banana
 Orange

2. ORDERED LIST

- An ordered list is created using the '' tag and each list item is defined with the '' tag
- The list items will be marked with numbers by default
- Use when Order Matters

Ex -

 Apple
 Banana
 Orange

3. Description list

<dl>

- ①
- ②
- ③

Example
<doctype html>

<html>

<head>

<title> My web page </title>

</head>

<body>

<h1> Welcome to my webpage, </h1>

<p> Here are some links: </p>

 Item 1

 Item 2

 Item 3

</body>

<h1> First Item </h1>

<h1> Second Item </h1>

<h1> Third Item </h1>

</body>

</html>

</html>

TABLES AND FRAMES IN HTML Lab Program-6

Both table and frames are used to organize and display content on web pages, but they serve very different purposes

HTML TABLES

• To display data in rows and columns, like spreadsheet or schedule.

• Tables are created using the <table> tag

• The table rows are defined with the <tr> tag

• The table cells are defined with the <td> tag

Tag

<table>

<tr>

<th>

<td>

Meaning

Starts a table

Table row

Table header cell (bold by default)

Table data cell

Lab Program-6
Working with tables

<table>

<thead>

<tr> table example </tr>

</thead>

<tbody>

<tr> border="1" </tr>

</tbody>

</table>

<tr> Month </tr>

<tr> savings </tr>

<tr>

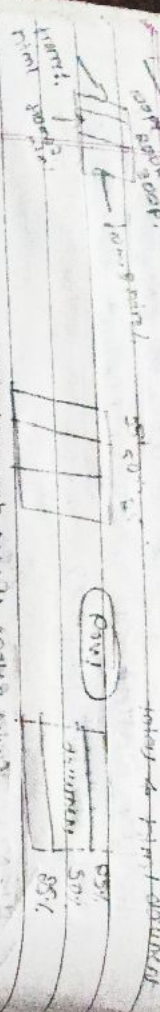
<td> January </td>

<td> \$100 </td>

</tr>

Month	Savings
January	\$100
February	\$800

rowspan, colspan



link → opening document
 (a) → opening document
 (b) → opening document

frame tag - src = attribute compulsory
 label tag - border tag compulsory
 align tag - align tag compulsory

along with href you should give target
 no attribute provided → opened open
 document attribute → document file

Lab program 8
 FRAMES → frames used when we have to add link

- Frames are used to divide a web page into multiple sections each section containing a separate HTML document
- To control the layout and size of these frames developers use framesets that allow for independent scrolling and customization of content

• The <frame> tag in HTML is used to define a single frame within a frameset.

- A <frameset> element is mainly used to create multiple frame layouts within a single page
- The attribute cols and rows are used to define the proportion in percentage or an absolute size value like pixels
- If you want to embed content from another page or site, use <iframe>.

ex - <iframe src = "https://google.com" width = "600" height = "400"></iframe>

The <Frameset> Tag attribute.

1. cols
 - a. Pixels
 - b. percentage
2. Rows
 1. Pixels
 2. percentage

1. COLS → vertical
- The cols attribute specifies the number of columns vertical frames in the frameset

• For each vertical frame the width can be set in two ways:

- a. Pixels → absolute
- For absolute width, the width is set in pixels, such as 100px or 800px

ex - set value of columns in a frame as
 cols = "100px, 200px, 300px"

example - <!DOCTYPE html>

<html>
<head>
<title> Frameset's col attribute in pixels </title>

</head>
<frameset cols = "100px, 200px, 300px">

<frame src = "frame3.html">

<frame src = "frame2.html">

<frame src = "frame3.html">

</frameset>

</html>

→ b. Percentage

- For relative width, the width is set as a percentage of the browser window such as 25%, 50% or 100%
- For example, set value of columns as a frame as `cols = "25%, 50%, 25%;"`

Example: - same as previous example but use % here

`<frame set cols = "25%, 50%, 25%;">`

2. Rows

horizontal

- The rows attribute specifies the number of rows (horizontal frames) in the frameset tag.
- For ex, to create three horizontal frame, set the value of rows = "25%, 50%, 25%;"
- It will create three rows, each occupying 25%, 50% and 25% of the browser window

Ex: - same as previous example but use

`<frameset rows = "25%, 50%, 25%;">`

WORKING WITH HYPERLINKS IN HTML

- Hyperlink in HTML used to link one page to another.
- Hyperlinks allows users to navigate b/w web pages open documents or trigger downloads
- In HTML hyperlinks are created using the `<a>` anchor tag, along with `href` attribute
- You can click on a link and jump to another document.
- When you move the mouse over a link, the mouse arrow will turn into a little hand.

Syntax..

`<a href = "URL" >link text </a>`

href

href → hyper text reference → the

Example 1:

Open link in another page

`<a href = "https://www.wikipedia.org" target = "_blank">`

Open Wikipedia

target = "_blank" opens the link in a new browser tab

2. link to another page in the same website

`<a href = "about:html" >about us </a>`

This links to a file named about.html in the same folder

3. link to a file for download

`<a href = "files/my-resume.pdf" >download </a>`

Download my Resume

The download attribute tells the browser to download the file instead of opening it

Example

```
<!DOCTYPE html>
<html>
<head>
<title>hyperlinks</title>
</head>
<body>
<a href = "https://www.wikipedia.org" target = "_blank">open wikipedia</a>
<a href = "about:html" >about us</a>
<a href = "files/my-resume.pdf" >download my app</a>
</body>
</html>
```


Audio = control src
2nd src = " "
type = " "
video = video src
2nd src = " " (video)

Lab 3 - The `img` element in HTML allows you to display images on a webpage by identifying their source.

- It uses attributes such as `src` and `alt`.

Example: - `img src = "image.jpg" alt = "description of`

ප්‍රකාශන

- 1911-1912

- Multimedia comes in many different formats
- It can be almost anything or have or give

- Polynomial

- The most common way to discover the type of a file is to look at the file extension

- Metilindia just have format and different extensions like :
 - .wav
 - .mpg

order. 60

Audio in HTML

- Videos in HTML

- `<video width="640" height="360" controls>`
- `<source src="video.mp4" type="video/mp4">`

- Poligamia

Useful Attributes and best practice

Tag	Attribute	Purpose
	alt	Screen readers
<audio> / <video>	controls	User playback controls
<video>	poster	Display image before playback
all	autoplay, loop, muted	Media behaviour control

1st

3

Description list in HTML

Description list is also known as delineation. It is used to display a list of terms and their corresponding description / meaning.

<dl> defines description list (delineation list)

<dt> defines a term in description list (delineation term)

<dd> defines delineation of term (delineation description)

ex -

<dl>

<dt> HTML LIDT

<dd> HTML stands for Hyper Text Markup Language LIDT

</dl>

defn. - A description list in HTML is a list that shows terms and their meaning & details.

dl → for the list

dt → for the term

dd → for the description

Lab Program - 3

Practicing use of multimedia components (image, video, audio in HTML documents)

<!DOCTYPE html>

<html>

<head>

<title> use of multimedia components </title>

</head>

<body>

<h1> multimedia components </h1>

<p> image example </p>

<p> video example </p>

<video example </p>

<video width = "640" height = "360" controls>

<source src = "https://www.w3schools.com/html/movie.mp4" type = "video/mp4">

</video>

<p>

</p>

<p> audio example </p>

<audio controls> <source src = "https://www.w3schools.com/html/horse.mp3" type = "audio/mp3">

</audio>

</body>

</html>

</html>

</html>

⑤ Designing of webpage working with list

```

<!DOCTYPE html>
<html>
<head>
<title> My web page: <title>
<head>
<body>
<h1> Welcome to my webpage: </h1>
<p> we are some list: </p>
<ul> type = "disc"
<li> apple </li>
<li> banana </li>
<li> orange </li>
</ul>
<ul type = "square"
<li> apple </li>
<li> banana </li>
<li> orange </li>
</ul>
<ul type = "circle"
<li> apple </li>
<li> banana </li>
<li> orange </li>
</ul>
<ul type = "none"
<li> apple </li>
<li> banana </li>
<li> orange </li>
</ul>

```

```

<ol type = "a"
<li> apple </li>
<li> banana </li>
<li> orange </li>
</ol>
<ol type = "A"
<li> apple </li>
<li> banana </li>
<li> orange </li>
</ol>
<ol type = "I"
<li> apple </li>
<li> banana </li>
<li> orange </li>
</ol>
<ol type = "I"
<li> apple </li>
<li> banana </li>
<li> orange </li>
</ol>

```


Lab Program - 01

Page No. _____
Date: / /

1) Acquaintance with elements, tags and basic structure of HTML file

<!DOCTYPE html>

<html>

<head>

<title> My first HTML Page </title>

</head>

<body>

<h1> Hello, world! </h1>

</p>

This is my first HTML program
 I am learning about tags and elements

</p>

<p> I st </p>

</p>

<h1> HTML </h1>

<h1> CSS </h1>

<h1> JavaScript </h1>

</p>

<p> visit my favourite website:

google

</p>

<p> Here is an image: </p>

</body>

</html>

Lab Program - 02

Page No. _____
Date: / /

<!DOCTYPE html>

<html>

<head>

<title> Text Formatting Practice </title>

</head>

<body>

<h1> Text Formatting in HTML </h1>

1. - Basic Formatting ->

<p> Bold text shows importance </p>

<p> Italic text is used for emphasis </p>

<p> <u> Underlined text </u> stands out visually </p>

<p> This is a <mark> highlighted </mark> word. </p>

<p> This is <small> smaller </small> text. </p>

<p> This is deleted text and

<ins> inserted </ins> text. </p>

<p> This is _{subscript} and ^{superscript} text. </p>

</body>

</html>

o/p pgm-02.

Hello, world!

This is my first HTML program

I'm learning about tags

and elements.

1st

• HTML

• CSS

• JavaScript

visit my favourite website: -

- google

Here is an image:

← If internet is not working → Sample image.

o/p pgm-2

Text Formatting in HTML

Bold text shows importance

Italic text is used for emphasis

Underlined text stands out visually

This is highlighted word

This is smaller text

This is deleted text and inserted text

This is subscript and superscript

text

Lab Program - 07

Page No.
Date / /

Practicing hyperlinking of web page

<!doctype html>

<html>

<head>

<title> My web page </title>

</head>

<body>

<p> link in another page </p>

<a href = "https://www.wikipedia.org"

target = "_blank"> wikipedia

<p> link in same page </p>

copy path

rose image

<p> link to file download </p>

pdf link /html-tutorial-pd1" download > download

"pdf path" download (pdf path in cpica

</body>

</html>

Lab Program - 08

Page No.
Date / /

Asst Designing of web page. working with frames

<!doctype html>

<html>

<head>

<title> My web page </title>

</head>

<frameset cols = "30%, 40%, 30%">

<frame src = "lab Program 1.html">

<frame src = "lab Program 2.html">

<frame src = "lab Program 3.html">

</frameset>

</html>

100%	100%	100%
2	3	3

- In HTML forms are used to collect user input and they consist of various controls like text fields, radio, buttons, checkboxes, dropdown menus and buttons

- These elements are grouped within a `<form>` tag which can send the collected data to a server using methods like GET or POST
- Controls are defined using tags like `<input>`, `<textarea>`, `<select>` and `<button>` each (customizable with attributes to handle a specific type of data entry and extraction)

Syntax:- `<form action="submit" -page.php" method="post">`

`<!-- form elements go here -->`

`</form>`

action: The URL where the form data should be sent

method: specifies how to send form data (GET or POST)

The `<input>` element

→ used to take input from user.

- HTML provides a variety of form controls (elements that allow user interaction):
- The HTML `<input>` element is the most used form element
- An `<input>` element can be displayed in many ways depending on the type attribute

HTML `<input>` types

- 1) `<input type="button">` → clickable button
- 2) `<input type="checkbox">` → select multiple items
- 3) `<input type="color">`
- 4) `<input type="date">`
- 5) `<input type="datetime-local">`
- 6) `<input type="email">` → email
- 7) `<input type="file">`
- 8) `<input type="hidden">` → store data which is not visible to user but sent to server when form is submitted.
- 9) `<input type="image">`
- 10) `<input type="month">`
- 11) `<input type="number">`
- 12) `<input type="password">`
- 13) `<input type="radio">` → select only one option.
- 14) `<input type="range">` min, max, value
- 15) `<input type="reset">` clears form
- 16) `<input type="search">` search operation
- 17) `<input type="submit">` submit form's data
- 18) `<input type="tel">` → mobile no.
- 19) `<input type="text">`
- 20) `<input type="time">`
- 21) `<input type="url">`
- 22) `<input type="week">`

Test Fields

- our - figure a number $\langle \text{text} \rangle$ defines a single line.

Example:-

Form 7

Label for = "name">|instance: L|label> dir

```
input_type = "xyz" pd = "name"
```

```
name = "name"><br>
```

or

$\langle p \rangle$ list Name: $\langle output \text{ type} = "text" \rangle \langle /p \rangle$

The Labell Elements → like acquisition, not different. Not held to

- The `label` tag defines a label for many forms

elmer's

- The for attribute of the `<label>` tag should be equal to the id attribute of the `<input>` element to bind them together.

② Radio Buttons

- The `Layout` type = "radio" > defines a radio button
- Radio buttons let a user select ONE of a limited number of choices.

Examply :-

Let users select one option from many

Charm

`<input type="radio" id="male" name="sex" value="male">`

$$V_{\text{eff}} = \frac{1}{2} \hbar^2 \omega_L^2$$

<body for="html">HTML</body>

☐
$$\angle \text{SSC}_1 = \text{right angle}$$

language

Label for "css" $\rightarrow$ css $\angle$ lbbd $\angle$ br $\angle$

3) Check Box

- The ϵ -input type "checkbox" defines a checkbox.
- checkboxes let a user select zero or more options of a limited number of choices

Example.

2011

-Logopus type = "checkbox" rd = "set:151"

name = "Whitley" value = "Bike 1"

$\angle \text{label} \text{ for} = \text{"white"} \angle \text{I have a circle} \angle \text{label} \angle \text{label} \angle \text{label}$

input-type = "checkbox" id = "checkbox1" style = "display: inline-block; vertical-align: middle; margin-right: 10px;">

norme = "veh(102)"

<label for="which">I have a color library

467

 $\angle \text{form}$

(+) The Submit Button

- The `<input type = "submit" >` defines a button for submitting the form data to a form handler.
- The form - handler is typically a file on the server with a script for processing input data
- The form - handler is specified in the form's action attribute

all -> input type = "submit" value = "submit")

⑤

The `<select>` element dropdown list

select elements during a drop-down list

ex:- Select

$\angle \text{option} \vee \text{value} \angle \text{!option}$

Option 7 bmw £100000

10p11on7 F9a+ 2 16p11on7

addition? Audi 210 option?

$\angle 1$ select

- the <option> element defines an option that can be selected
- By default, the first item in the drop-down list is selected
- To define a pre-selected option add the selected attribute to the option

Input Type Password

→ type="password" attribute are hidden

5. <input type="password"> defines a password field

Ex:- <label for="pwd">Password: </label>
 <input type="password" id="pwd" name="pwd"> </input>

Input Type Button

<input type="button"> defines a button.

Ex:- <input type="button" onclick="alert('Hello world!')" value="Click" />

Input type color → choose color

• The <input type="color"> is used for input fields that should contain a color

• Depending on browser support, a color picker can show up in the input field

Ex:- <label for="jovcolor">Select your favorite color: </label>
 <input type="color" id="jovcolor" name="jovcolor" value="#100000">

Select your favourite color



①

<!DOCTYPE html>

<html>

<head>

<title> html forms </title>

</head>

<body>

<form>

<p> STUDENT INFORMATION </p>

<!-- Student Name -->

<label for = "s"> Student Name: </label>

<p> student Name: <input type = "text" id = "s" name = "s" placeholder = "enter your name" = "s" required>

<!-- Mobile number -->

<p> Mobile no: <input type = "text" id = "m" name = "m" placeholder = "enter your mobile no" = "m" required>

<!-- Email ID -->

<p> email id: <input type = "email" id = "e" name = "e" placeholder = "enter your email" = "e" required>

<!-- Father Name -->

<p> Father name: <input type = "text" id = "f" name = "f" placeholder = "enter your father name" = "f" required </p>

<!-- Mother Name -->

<p> mother name <input type = "text" id = "mo" name = "mo" placeholder = "enter your mother name" = "mo" required </p>

<!-- course selection -->

<p> select course </p>

<select id = "course" name = "course" required>

<option value = " " > -- select -- </option>

<option value = "Bca" > Bca </option>

<option value = "Bsc" > Bsc </option>

<option value = "Bcom" > Bcom </option>

</select>

<!-- Date of Birth -->

<p> Date of Birth: <input type = "date" id = "dob" name = "dob" required > </p>

<!-- Gender -->

<p> Gender: <input type = "radio" id = "male" name = "gender" value = "Male" required

<input type = "radio" id = "female" name = "gender" value = "Female" >

Female

</p>

<!-- Photo upload -->

<p> Student Photo: <input type = "file" id = "photo" name = "photo" accept = "image/*" required > </p>

</p>

<!-- Branch selection -->

<p> select Branch:

<input type = "checkbox" id = "cse" name = "branch"

value = "cse" > CSE

<input type = "checkbox" id = "ece" name = "branch"

value = "ece" > ECE

<input type = "checkbox" id = "civil" name = "branch"

value = "civil" > Civil

</p>

<!-- Address -->

<p> Address:

<textarea id = "address" name = "address"

rows = "4" cols = "40" </textarea>

</p>

<!-- Submit -->

<input type = "submit" value = "Submit">

</form>

</body>

</html>

Introduction

- CSS, which stands for Cascading Style Sheets, is a style sheet language used for describing the presentation and visual appearance of HTML (and XML) documents.
- It provides a way to separate the structure and content of a web page from its design and layout.
- CSS allows you to define various styles and rules that control how elements are displayed on a web page.
 - ① controls look & feel of web page - range colors, fonts, spacing

CSS Syntax ② separates content & design
 → HTML for structure
 CSS for styling
 → makes webpage attractive & friendly

- CSS rules consist of a selector and a declaration block.

- Selector { Property 1: value 1; Property 2: value 2;; }

Diagram illustrating the CSS rule structure:

```

  selector      declaration      declaration
  |-----|-----|-----|
  h1 { color : red ; font - size ; 14px ; }
  ↑         ↑         ↑         ↑         ↑
  Selector  property value property value
  
```

- **Selector** :- Selector indicates the HTML element you want to style.

It could be any tag like <h1>, <title> etc

- **Declaration Block** :- The declaration block can contain one or more declarations separated by a semicolon.

For the above example, there are 2 declarations:

1. color : red;

2. font-size : 14px;

- Each declaration contains a property name & value separated by a colon.

- **Property** :- A property is a type of attribute of HTML element. It could be color, border etc

value: value are assigned to CSS properties in the above example value "red" is assigned to color property.

LEVELS OF STYLE SHEET OR TYPES OF CSS

- CSS is added to HTML pages to format the document according to information in the style sheet.

Types

There are 3 ways to insert CSS in HTML documents.

① Inline CSS

② Internal CSS

③ External CSS

①

INLINE CSS

- Inline CSS is used to apply CSS on a single line or element.

Syntax - `<htmltag style="css property1: value; css property2: value;" > </htmltag>`

Ex - `<p style="color: blue;" > Hello CSS </p>`

②

INTERNAL CSS

- Internal CSS is used to apply CSS on a single document or page.
- It can affect all the elements on the page.
- It is written inside the style tag within head section of HTML.

`<? <head>`

`<style>`

`p {color: blue;}`

`</style>`

`</head>`

`<body>`

`<p> --- </p>`

`<p> --- </p>`

`</body>`

③

EXTERNAL CSS

- External CSS is used to apply CSS on multiple pages or on a page.
- Here we write all the CSS code in a CSS file.
- File extension must be .css
- Ex - style.css

`p {color: blue;}`

You need to link this style.css to your HTML page like this

`<head>`

`<link`

`rel="stylesheet" href="style.css">`

`</head>`

CSS SELECTOR

Page No.
Date

• CSS selectors are used to select the content you want to style.

• Selectors are the part of CSS rule set.

• CSS selectors select HTML elements according to their id, class, type, attribute etc.

• There are several different types of selectors in CSS.

• 1. CSS Element selector

2. CSS ID selector

3. CSS class selector

4. CSS Universal selector

5. CSS group selector.

⇒ attribute selector

① CSS ELEMENT SELECTOR

• The element selector selects the HTML element by name.

<html>

<head>

<style>

p

h

text-align: center;

color: blue;

{

</style>

</head>

<body>

<p> This style will be applied on every paragraph <p>

</body>

</html>

② CSS ID SELECTOR

• The id selector selects the id attribute of an HTML element to select a specific element.

• An id always unique within the page. So it is chosen to select a single, unique element.

• It is written with the hash character (#) followed by the id of the element.

<head>

<style>

#para1 {

text-align: center;

color: blue;

{

</style>

</head>

<body>

<p id="para1"> I am ID Selector </p>

</body>

③ CSS CLASS SELECTOR

• The class selector selects HTML elements with a specific class attribute.

• It is used with a period character. (full stop symbol) followed by the class name.

<head>

<style>

.center {

text-align: center;

color: blue;

{

</style>

</head>


```

<body>
<h1 class="center"> this heading is blue and
text-align: center;
</body>

```

④ CSS UNIVERSAL SELECTOR

The universal selector is used as a wildcard character. It selects all the elements on the page.

```

<html>
<style>
* {
color: green;
font-size: 20px;
}
</style>
</html>

```

```

<h2> This is heading </h2>
<p> this style will be applied on every paragraph </p>
<p id="para1"> Me too! </p>
</body>

```

⑤ CSS GROUP SELECTOR

- The grouping selector is used to select all the elements with the same style definitions
- Grouping selector is used to minimize the code commas are used to separate each selector in grouping

```

<html>
<style>
h1, h2, p {
text-align: center;
color: blue;
}
</style>
</html>
<body>
<h1> Hello Travelpoint.com </h1>
<h2> Hello Travelpoint.com (30 smaller font) </h2>
<p> this is a paragraph </p>
</body>

```

CSS VALUES

- Values are written immediately after the color to separate them from CSS property
- CSS values aren't just text, they come in different forms - URLs, units, integers, strings, inherit, auto, none etc

① Text

- Text values are prevalent (most commonly used) in
 - They are usually written without quotes unlike string
 - Every valid CSS property has a text or a value
- ```

h1 {
color: red;
text-align: center;
}

```
- Red and center are text values of color and text-align



## ② INTEGER

- Integers are numbers from 0-9.
- Some CSS properties make allowance for their values to be defined as integers, even negative integers.
- The values for properties like column-count and z-index are defined using integers.

div {

column-count: 4;

div {

z-index: -1;

## ③ URLS

- Properties like background-image take an actual URL as their value.
- This URL could be absolute or relative.
- Meaning you could copy the address of an image online and use it as your background image or you could get images from the project directory folder you are working with on your computer.

div {

background-image: url

url('https://cdn.pixabay.com/photo/2015/10/12/29/20/20/11/340.jpg')

div {

background-image: url('img1.jpg');

## ④ STRINGS

- Strings are text within quotes.
- The content property which you can use to insert generated content allows strings within its parameter.

div {

background-color: #f0f0f0;

background-color: #f0f0f0;

background-color: #f0f0f0;

background-color: #f0f0f0;

background-color: #f0f0f0;

background-color: #f0f0f0;

background-color: #f0f0f0;

background-color: #f0f0f0;

background-color: #f0f0f0;

background-color: #f0f0f0;

background-color: #f0f0f0;

background-color: #f0f0f0;

## ⑤ COLORS

- The color / background-colors are properties used to set the color of HTML elements.
- Their values accept either a hexadecimal color combination value, an RGB specification or a named color.

div {

background-color: red;

div {



```
div {
background-color: #000000;
}
```

⑥ UNITS  
css must be used to position items on web page, general layout, and media queries, many properties take units and measurements as their values.

```
h3 {
font-size: 24px;
}
```

⑦ AUTO  
Auto values allows the property to fill or adjust according to the content of the element.

```
div {
width: auto;
margin: auto;
}
```

⑧ NONE  
None adds no value to the specified property.

```
div {
border: none;
}
```

⑨ INHERIT  
Inherit keyword allows the property to inherit same value from the parent element.

- All CSS properties accept inherit.

```
<div>
```

```
<p> this is me </p>
```

```
<p id="prime"> this is not me </p>
```

```
</div>
```

```
prime {
color: red;
}
```

```
#prime {
color: inherit;
}
```

THE SPAN AND DIV TAGS

The `<span>` and `<div>` tags are two commonly used HTML tags for grouping and styling elements.

① `<span>` TAG

The `<span>` tag is an inline-level HTML element that is used to group and apply styles to inline elements or small sections of text within a larger block of content.

The `<span>` tag is often used in combination with CSS to target and style specific parts of text or inline elements.

Example:

```
<div>
```

```
<html>
```

```
<body>
```

```
<p> This is an inline span this word element inside a paragraph. </p>
```

```
<p> The span element is an inline element and will not start on a new line and only takes up as much width as necessary </p>
```



</body>  
</html>

## ② <div> Tag

- The <div> tag is a block-level HTML element that is used to group and structure larger sections of content on a web page.
- It is a generic container and doesn't carry any semantic meaning on its own.
- The <div> tag is commonly used for layout purposes, providing a way to divide and organize different sections of a web page.

Ex  
<doctype html>

<html>

<body>

<p style="border: 1px solid black">Hello world

<div style="border: 1px solid black">Hello world

<p> The p and the div elements are both block

elements, and they will always start on a new line and take up the full width available (stretch out to the left and right as far as it can). </p>

</body>

</html>

## CSS COMMENTS

- CSS comments are generally written to explain your code.
- It is very helpful for the users who read your code so that they can easily understand the code.
- Comments are ignored by browsers.
- Comments are single or multiple line statements and written within /\* ... \*/

Example any ex

## CSS PROPERTIES

- CSS properties are the styles used on specified selectors.
  - They are written before values in the CSS rule set and are separated from property values by colon.
  - Explain these selectors and elements have different property. Some property are universal and can be used on every selector.
  - Others work only on specific selectors and under particular conditions.
  - There are 4 common properties to work
- ① Text property
  - ② Font property
  - ③ Border property
  - ④ List property
- Background property  
Margin property  
padding property



## I) TEXT PROPERTIES

Property	Description	Value
① color	sets the color of a text	Hex, RGB, Keyword
② text-transform	sets the capitalization of the text	uppercase, lowercase, capitalize
③ text-align	sets the alignment of the text on the screen	right, left, center, justify, stretch, text-align
④ letter-spacing	sets the spacing b/w text characters	normal, length, $n$ px
⑤ text-decoration	sets the decoration added to the text	none, underline, line-through, overline
⑥ text-shadow	add shadow effect to text	horizontal, vertical, color

### COLOR

In CSS we can control the color of text using the color property. There are different ways to specify colors

→ ① Named colors (Keywords)

CSS provides a set of predefined colors that can used directly

ex: color: red; grey, lightblue

red, green, blue, yellow, orange,

→ ② hexadecimal color

hexadecimal code represent colors using a combination of 6 alpha numeric characters (0-9 to A-F)

ex: color: #1a3a56

→ ③ RGB colors

RGB value represent colors using Red, Green, Blue color channels.

Range is 0 to 255

ex: color: rgb(0, 1, 2)

using hex

## Font Properties

ex: Program for text property

<!DOCTYPE html>

<html>

<body>

<p style="border: 1px solid black">Hello world

</p>

</body>

</html>

<pre> </pre>

<div>

<div>

<div>

<div>

<div>

<div>

<div>

<div>

<div>

<div>

<div>



FONT PROPERTY ES

Property	Description	Value
① font-family	Specifies the font or a list of fonts to be used. The browser will try to use the first font in the list that is available in the user's system.	Arial, sans-serif, monospace.
② font-weight	Sets the weight of a font.	100 - Thin (light), 200 - Extra light, 300 - Medium, 400 - Normal, 500 - Medium, 600 - Semi bold, 700 - Bold, 800 - Extra bold, 900 - Black (Heavy)
③ font-style	Sets the style of a font.	normal, italic, oblique
④ font-variant	Specifies whether text should be displayed in a small caps font.	small-caps
⑤ font-size	Controls the size of text.	12pt (or) provide precise control over font size



font: italic small-caps 10px; color: red;

**SHORTHAND PROPERTY**  
 Font: font-style, font-variant, font-weight, font-size/line-height, font-family

Example Program for FONT PROPERTIES  
 <!DOCTYPE html>

<html>

<head>

<style>

div { border: 1px solid grey; }

h1 {

font-style: Normal;  
 font-variant: small caps;

}

p {

font-weight: bold;

font-size: 14px;

font-family: Arial;

}

text formatting  
 this is typed  
 with -----

</style>

</head>

<body>

<div>

<h1> This is example for FONT PROPERTIES  
 <p> This is example program (onset of

Font properties such as font-weight,  
 font-size and font-family etc </p>

border: border: width, style, color.

## BORDER PROPERTIES

• background color  
 • border: 1px solid black  
 • border: 1px solid red  
 • border: 1px solid blue  
 • border: 1px solid green  
 • border: 1px solid yellow  
 • border: 1px solid cyan  
 • border: 1px solid magenta  
 • border: 1px solid black  
 • border: 1px solid red  
 • border: 1px solid blue  
 • border: 1px solid green  
 • border: 1px solid yellow  
 • border: 1px solid cyan  
 • border: 1px solid magenta

Property	Description	Values
border	Sets the shorthand combination for border width, border-style and border-color	border-width, border-style, border-color
border-width	Sets the thickness of the border	values can be specified in units like pixels (px), em, medium, thick, thin
border-color	Sets the color for the border	keyword, RGB, hex, transparent
border-style	Sets the style for an element's border	none, hidden, dotted, solid, dashed, double

Write a Program on Border Property

<!DOCTYPE html>

<html>

<head>

<p style="border: 1px dashed red"> Border Property

</style>

h1 {

border-style: solid;

border-color: red;

border-width: thick;

}

shorthand  
 border: 1px solid red;



P. 8

border-style: thin  
border-color: black

[border: solid red 2px;

shorthand  
↑ style color width

<style>

</head>

<body>

<div border="1" style="border: 1px solid black;">

</div> this is styled with border properties

</body>

</html>

border

this is style & width

same border properties

## LIST PROPERTY

| Property            | Description                                                                           | Value                                                                                                          |
|---------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| list-style-type     | short-hand combination for list-style-type, list-style-position, and list-style-image | list-style-type, list-style-position, list-style-image                                                         |
| list-style-position | specifies the position of the list-item markers                                       | inside, outside                                                                                                |
| list-style-type     | sets an image as the list-item marker                                                 | none, url                                                                                                      |
| list-style-type     | This sets the type of list-item marker                                                | Property controls the marker (e.g circle, square, decimal, lower-alpha, lower-roman, upper-alpha, upper-roman) |

① list-style-type

② list-style-position

③ list-style-image

list-style: square inside none

list-style: square inside none

ex. - lab program 13



## Date \_\_\_\_\_

Page No.

## Date: \_\_\_\_\_

2180CTYPE hlm17

210019

7 mod

Forms  $\angle f i t k u \rangle$

2 heads

1 term

<label>First name: </label>

Looped type = "text" > "bdi" < />

Label > last name: Label >

$$\text{limp\_type} = \text{"let"} \geq \text{let} \rightarrow$$

Age at birth:  $\angle$  (label)

[illegible]

input type = "radio" name = "gender"

$$\angle abd > \angle mde < \angle abd$$

let input\_type = "radio" name = "gender" >

20017 female 4/11/01 7 48617 1/1/01

7. email id: cllobu27@gmail.com

$$\angle \text{out} + \mu p = \angle \text{in} + \mu r$$

Labelled axoname: 2/100077

$\angle \text{input type} = \angle a + \angle b + \angle c = 180^\circ$

label's primary password: <label>

$\text{tropus } + \text{gr} = \text{parabola} \quad ( \text{ } \angle \text{dr } )$

Labels (course & library)

|                                                                                                               |      |
|---------------------------------------------------------------------------------------------------------------|------|
| $\angle$ seen /<br>$\angle$ option $\rightarrow \dots$ seen $\rightarrow \dots$ $\angle$ option $\rightarrow$ | 100% |
|---------------------------------------------------------------------------------------------------------------|------|

Option 1:  $\text{BR} < \text{Option 2}$

Capillary blood  $< 1 \text{ mm}$   
 $< 1 \text{ second} > 1 \text{ hr}$

Label > certification Label / End /

```
input type = "checkbox" /
```



## ① Inline CSS

- Inline CSS is used to apply CSS on a single line
- Inline CSS is written inside the HTML <sup>body</sup> tag using style attribute
- It affects only that specific element which is mentioned
- Inline CSS has highest priority over other CSS types.

Ex - `<!DOCTYPE html>` • note reusable, makes HTML code length

```

<html>
<head>
<title> Inline CSS example </title>
</head>
<body>
<h1 style = "color: blue; text-align: center;" >
This is inline CSS </h1>
<p style = "font-size: 18px; color: green;" >
This is paragraph with inline CSS </p>
</body>
</html>

```

## ② Internal CSS

- Internal CSS is used to apply CSS on a single document or page.
- It can affect all the elements of the page
- It is written inside the `<style>` tag within head section of HTML.

Ex `<!DOCTYPE html>`

```

<html>
<head>
<title> Internal CSS </title>
<style>
body {
background-color: blue;
}
h1 {
color: darkred;
text-align: center;
}

```



```

p {
 color: darkblue;
 font-size: 18px;
}

```

```

</style>
</head>
<body>

```

```

<h1> This is internal CSS </h1>
<p> This paragraph is styled using internal
 CSS </p>

```

```

</body>
</html>

```

### ③ External CSS

- External CSS is used to apply CSS on multiple page or all page
- The we write all the CSS code in a CSS file
- Its extension must be .css
- External CSS is written in a separate .css file and linked to the HTML file using the <link> tag, it is easy to maintain.

HTML file: index.html

```

<!DOCTYPE html>
<html>

```

```

 <head>

```

```

 <link> external CSS example </link>

```

```

 <link rel = "stylesheet" type = "text/css"

```

```

 href = "style.css" >

```

```

 </head>

```

```

 <body>

```

```

 <h1> This is external CSS </h1>

```

```

 </body>
 </html>

```

```

</body>
</html>

```

css file :- style.css

```

body {

```

```

 background-color: lightyellow;
}

```

```

h1 {

```

```

 color: purple;
 text-align: center;
}

```

```

p {

```

```

 color: red;
 font-size: 18px;
}

```

```

 </body>
 </html>

```

```

 </body>
 </html>

```

```

 </body>
 </html>

```

```

 </body>
 </html>

```

```

 </body>
 </html>

```

```

 </body>
 </html>

```

```

 </body>
 </html>

```

```

 </body>
 </html>

```

```

 </body>
 </html>

```

```

 </body>
 </html>

```

```

 </body>
 </html>

```



### Lab Program - 10

(a) acquaintance with creating stylesheet, CSS properties and styling

```

<!doctype html>
<html>
<head>
<title> internal CSS </title>
<style>
body {
background-color: blue; color: white;
h1 {
color: darkred;
text-align: center;
p {
color: darkblue;
font-size: 18px;
}
}
</style>
</head>
<body>
<h1> This is internal CSS </h1>
<p> This paragraph is styled using internal CSS </p>
</body>
</html>

```

### Lab Program - 12

(a) working with list property

```

<!doctype html>
<html>
<head>
<style>
#p {
list-style-type: disc outside none;
#b {
list-style-type: circle outside none;
#c {
list-style-type: square outside none;
#d {
list-style-type: decimal inside none;
#e {
list-style-type: lower-alpha inside none;
#f {
list-style-type: upper-alpha inside none;
}
}
</style>
</head>
<body>
<ul id="a">
 Apple
 Banana
 Orange


```



## CSS Background Property

- \* The CSS background is the area behind an element's content which can be color, image or both.
- \* The background property lets you control these aspects including color, image, position and repetition.

| Background              | Description                                                                         | Values                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| ① Background-color      | Defines the background color of an element.                                         | color name, hex or RGB values                                                                            |
| ② Background-image      | Adds one or more images as the background of element.                               | url ("path")<br>(with "path" inside url)                                                                 |
| ③ Background-repeat     | Specifies how the background image should be repeated.                              | Repeat, repeat-x, repeat-y<br>Repeat: image<br>Repeat-x: horizontal<br>Repeat-y: vertically<br>no-repeat |
| ④ Background-position   | Determines the initial position of the background image within the element.         | Left-top, left-center, left-bottom, top, right-bottom, center-top, center, center-bottom, bottom         |
| ⑤ Background-attachment | Property sets whether a background image scrolls with the rest of page or is fixed. | scroll, fixed                                                                                            |
| ⑥ Background-size       | Property specifies the size of the background image.                                | length(px), percentage                                                                                   |

2. id = "b">

<li> parrot </li>

<li> peacock </li>

<li> eagle </li>

</ul>

2. id = "c">

<li> lion </li>

<li> monkey </li>

<li> tiger </li>

</ul>

2. id = "d">

<li> apple </li>

<li> banana </li>

<li> orange </li>

</ul>

2. id = "e">

<li> parrot </li>

<li> peacock </li>

<li> eagle </li>

</ul>

o/p.

</body>

</html>

• apple

• banana

• orange

• parrot

• peacock

• eagle

• lion

• monkey

• tiger

• apple

• banana

• orange

• parrot

• peacock

• eagle

• lion

• monkey

• tiger

• apple

• banana



## CSS MARGINS

\* The CSS margin property is used to create a space around elements, outside of any defined borders.

- \* Margins define the distance between an element's border and the surrounding elements.
- \* CSS has properties for setting the margin for each individual side of an element (top, right, bottom, and left) and a shorthand property for setting all the margins property in one declaration.

### Margin Property

- \* `margin-top`: sets the top margin of an element
- \* `margin-right`: sets the right margin of an element
- \* `margin-bottom`: sets the bottom margin of an element
- \* `margin-left`: sets the left margin of an element
- \* Shorthand property: - `margin: top right bottom left`
- \* Margin Property Values
  - \* `auto` - the browser calculates the margin
  - \* `length` - specifies a margin in px
  - \* `%` - specifies a margin in % of the width of the containing element
  - \* `inherit` - specifies that the margin should be inherited from the parent element.

### Example

`<!doctype html>`

`<html>`

`<style>`

`div {`

`border: 1px solid black;`

`margin-top: 100px;`

`margin-bottom: 100px;`

`margin-right: 150px;`

`margin-left: 80px;`

`background-color: lightblue;`

`</style>`

`</html>`

`<body>`

`<div>`

`</div>`

`</body>`

`</html>`

`</div>`

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`</body>`



Ex:- 2. Doctype html7

```
<html>
<head>
<style>
```

```
div {
border: 1px solid black;
```

```
background-color: lightblue;
```

```
padding-top: 50px;
```

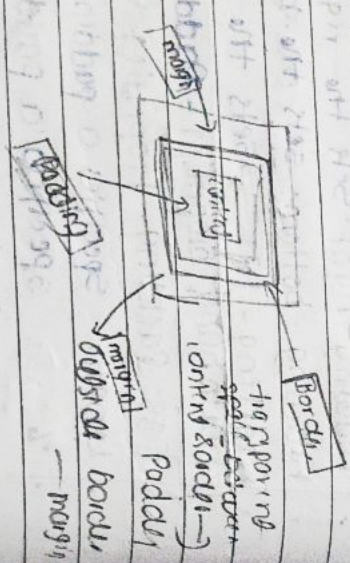
```
padding-right: 30px;
```

```
padding-bottom: 50px;
```

```
padding-left: 80px;
```

```
</style>
</head>
```

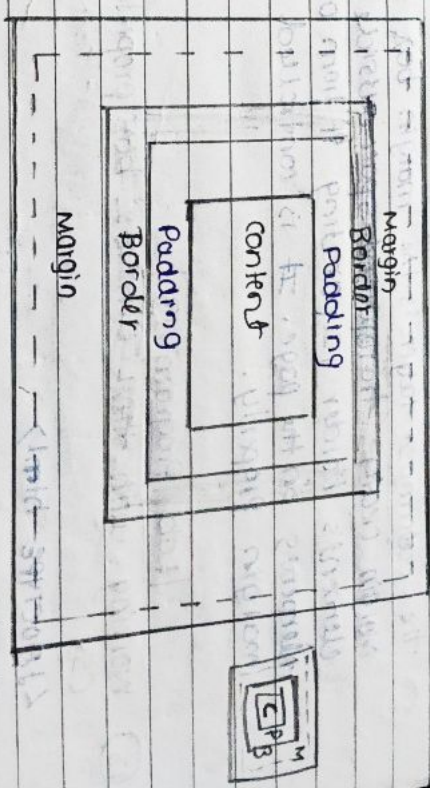
<body>  
<div> This div element has a top padding of 50px, a right padding of 30px, a bottom padding of 50px, and a left padding of 80px </div>



## CSS Box Model

- The CSS Box Model is a fundamental concept in web development that describes how HTML elements are structured and how their dimensions are calculated and displayed on a webpage.
- Every HTML element is treated as a rectangular box.
- Every box consists of 4 parts
  - ① content
  - ② padding
  - ③ borders
  - ④ margins

The image below illustrates the CSS Box Model



### ① Content Box

This is the innermost area where the actual content of the element resides, such as text, images, or other HTML elements. Its dimensions are typically defined by the width and height properties. In CSS



② Padding Box - The innermost box is the padding surrounding the content box inside the box. Padding creates transparent space inside the element between the content and the border. It is controlled by the padding property.

③ Border Box - The border box encloses both the content and padding. It represents the visual border of the element, which can be styled with properties like border-width, border-style and border-color.

④ Margin Box - The outermost layer is the margin box. The margin creates transparent space outside the element's border, separating it from other elements on the page. It is controlled by the margin property.

Lab Program - 13

⑬ Working with HTML elements box properties in CSS

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
border: 1px solid red;
padding: 10px;
margin: 10px;
}
```

```
</style>
</html>
```

```
<head>
<body>
<div> This is a program explaining the
box properties such as border, padding,
margin </div>
```

Lab Program - 15

⑮ Designing with a cascading style sheet (CSS)

```
<!DOCTYPE html>
<html>
<head>
<style>
```

```
body {
background-color: blue;
margin: 10px;
text-align: center;
}
```

```
h1 {
color: black;
font-size: 24px;
}
```

```
div {
border: 1px solid red;
padding: 10px;
margin: 10px;
}
```



```

</style>
</head>
<body>
 <h1> This is a CSS program </h1>
 <p> This is a program that consists CSS
 properly. Internal CSS </p>

```

<div> This is a div tag </div>

```

</body>
</html>

```

### Lab Program - 11

(11) working with background, text and font properly.

```

<!doctype html>

```

```

<html>

```

```

<head>

```

```

<style>

```

```

 h1 {

```

```

 color: red;

```

```

 text-transform: uppercase;

```

```

 text-align: center;

```

```

 }

```

```

 p { font-family: Arial;

```

```

 font-style: normal;

```

```

 font-size: 14px;

```

```

 }

```

```

 body { background-color: lightblue;

```

```

 background-size: 40px 50px;

```

```

 background-position: center;

```

```

 background-image: url('');

```

```

 }

```

```

 background-repeat: no-repeat;

```

```

}

```

```

</style>

```

```

</body>
 <h1> This is a CSS program </h1>
 <p> This program contains background,
 text, font </p>

```

```

</body>
</html>

```

### Lab Program - 16

(16) working with Designing with cascading styled - external CSS

```

<!doctype html>

```

```

<html>

```

```

<head>

```

```

 <title> External CSS </title>

```

```

 <link href="style.css" rel="stylesheet" type="text/css">

```

```

</head>

```

```

<body>

```

```

 <h1> This is external CSS </h1>

```

```

 <p> This is paragraph </p>

```

```

</body>

```

```

</html>

```

Save as style.css

```

body {

```

```

 background-color: lightblue;

```

```

 background-size: 40px 50px;

```

```

 background-position: center;

```

```

 background-image: url('');

```

```

 background-repeat: no-repeat;

```

```

 text-align: center;

```

```

}

```

```

p { color: red;

```

```

 font-size: 14px;

```

```

}

```

This is internal CSS

Internal CSS

Paragraph



## CSS DIMENSIONS

- CSS dimensions refer to the property used to control the size and spacing of HTML elements on a web page.
- These properties are crucial for defining layouts and ensuring elements occupy the desired amount of space.

### Property

- width and height: These properties directly set the width and height of an element's content area.
- margin: This property sets the space around an element.
- padding: This property sets the space between an element and its container.
- border: This property sets the width, height, and style of the border around an element.
- max-width & max-height: These properties set the maximum width and height an element can have.
- min-width & min-height: These properties define the minimum width and height of an element.

### Write a program of CSS dimensions

```
<!doctype html>
<html>
<head>
<title> CSS DIMENSIONS </title>
<style>
```

```
body {
background-color: lightblue;
```

```
max-width: 4px;
```

```
max-height: 5px;
```

```
height: 80px;
```

```
width: 100px; 100px;
```

```
}
```

```

p {
min-width: 15px;
```

```

min-height: 10px;
}
```

```
</style>
```

```
</head>
```

```
</body>
```

```
</html>
```

```

<!-- CSS DIMENSIONS -->
<!-- This is a program including CSS
Dimension <!-- -->

```

```

<body>
<div>

```

```

</div>
</body>

```

```

</div>

```

```

</div>

```

```

</div>

```

```

</div>

```

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```

</div>

```

```

</div>

```

```

</div>

```



create a class TimeTable

<!DOCTYPE html>

<html>

<head>

<title> class time table </title>

<style>

table {

border-collapse: collapse;

width: 80%;

margin: 50px auto;

text-align: center;

}

tr, td {

border: 2px solid black;

padding: 10px;

}

tr {

background-color: white;

}

p {

text-align: center;

font-size: 22px;

font-weight: bold;

margin-bottom: 10px;

}

>

class TimeTable </p>

</>

<tr> Day </tr>

<tr> 9:30 - 10:30 </tr>

<tr> 10:30 - 11:30 </tr>

<tr> 11:30 - 11:45 </tr>

<tr> 11:45 - 12:45 </tr>

</table>

<tr> Monday </tr>

<td> Accounting </td>

<td> English </td>

<td rowspan="6"> Break </td>

<td> Web Tech </td>

<td> Java </td>

<td rowspan="6"> Break </td>

</tr>

<tr>

<td> Tuesday </td>

<td> English </td>

<td> Web Tech </td>

<td> Accounting </td>

<td> Java Lab </td>

</tr>

<tr>

<td> Wednesday </td>

<td> Java </td>

<td> Java Lab </td>

<td> Web Tech </td>

<td> Accounting </td>

</tr>

<tr>

<td>

<td> Thursday </td>

<td> Accounting </td>

<td> Web Tech </td>

<td> Java Lab </td>

<td> Kannada </td>

</tr>

<tr>

<td> Friday </td>

<td> English </td>

<td> Accounting </td>

<td> Java </td>

<td> Kannada </td>

</tr>

<tr>

<td>

<td>

<td>

<td>

<td>



# Class Time Table

| Day       | 9:30 - 10:30 | 10:30 - 11:30 | 11:30 - 11:45 | 11:45 - 12:45 | 12:45 - 1:30 |
|-----------|--------------|---------------|---------------|---------------|--------------|
| Monday    | Accounting   | English       |               | Web Tech      | Java         |
| Tuesday   | English      | Web Tech      |               | Accounting    | Java Lab     |
| Wednesday | Java         | Java Lab      | Break         | Web Tech      | Accounting   |
| Thursday  | Accounting   | Web Tech      |               | Java Lab      | Kannada      |
| Friday    | English      | Accounting    |               | Java          | Kannada      |
| Saturday  | Kannada      | Web Tech      | Java          | Break         |              |

<:doctype html>

<body>

<body> doat over <

<body> printout <

<body> history <

<body> design <

<body> probout <



## CSS DISPLAY

- The
- The display property is an important CSS property for controlling layout
- It specifies whether an HTML element is treated as a block or an inline element
- Every HTML element has a default display value depending on what type of element it is.
- The default display value for most elements is block or inline.
- The display property is used to change the default display behavior of HTML elements

### Property

#### 1) DISPLAY: BLOCK

- Elements with display: block; are rendered as block-level elements.
- They create a line break after the element and occupy the full width of their parent container.
- This is the default property for <div> elements
- It places them vertically, one after another
- You can adjust the height and width of a block level element.

<div><p><h1>

#### 2) DISPLAY: INLINE

- Use this property to display an element inline.
- Elements with display: inline; are rendered as inline elements
- They don't create line breaks and occupy only the necessary width & height based on their content
- It does not start a new line and respects the content flow

#### 3) DISPLAY: INLINE-BLOCK

- Elements with display: inline-block; are rendered as inline-level block elements
- They behave like inline elements but allow you to set width, height, padding and margins.
- Combining characteristics of both block and inline, this value allows element to flow inline while still having block-level properties
- It's useful for creating responsive layouts

#### 4) DISPLAY: NONE → hide

- Elements with display: none; are not rendered and are completely hidden from the page
- They don't take up any space in the document flow.
- This property hides the div or the container which we use this property
- Using it on one of the div it will make working clear

#### Example

<!DOCTYPE html>

<html>

<head>

<title> DISPLAY </title>

<style>

#a {

height: 100px;

width: 200px;

background: teal;

display: block;

}

#b {

height: 100px;

width: 200px;

background: cyan;

display: block;

}



```
#c {
 height : 50px;
 width : 200px;
 background : green;
 display : inline;
}
```

```
</table>
</head>
<body>
 <div>
 <div id = "a"> Block 1 </div>
 <div id = "b"> Block 2 </div>
 <div id = "c"> Block 3 </div>
 </div>
</body>
</html>
```

o/p.



Float →  
newspaper  
like newspaper

left T  
initial T

right

## CSS FLOAT PROPERTY

- The CSS float property is used to move an element out of the normal document flow and position it to the left or right of its container.
- moving element left or right out of normal document flow

### FLOAT VALUES

| Value     | Description                                                                                       |
|-----------|---------------------------------------------------------------------------------------------------|
| ① left    | Element floats on the left side of the container, allowing content to flow around its right side. |
| ② right   | Element floats on the right side of the container, allowing content to flow around its left side. |
| ③ initial | Element is set to its default value.                                                              |
| ④ inherit | The element inherits the floating property from its parent element.                               |

### Example Program for Float

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
span {
```

```
 float : left;
```

```
 width : 100px;
```

```
 font-size : 400%;
```

```
 font-family : times new roman;
```

```
 line-height : 80%;
```

```
}
```

```
</style>
```

```
</head>
```



<body>

<p> <span>T </span> HTS is some text.

This is some text. This is some text.  
This is some text. This is some text.

and contents

</p>

<p> In the paragraph above the first letter of  
the text is embedded in the span element

</p>

</body>

</html>

O/P.

This is some text

In the paragraph

## CSS Pseudo-classes - style element based on state

- \* A CSS pseudo-class is a keyword is a keyword that can be added to a selector, to define a style for a special state of any element.
- \* Some common use for pseudo-classes:-
  - style an element when a user moves the mouse over it.
  - style visited and unvisited links differently
  - style an element when it gets focus.
  - style valid/invalid/required/optional form elements
  - style an element that is the first child of its parent.

### Syntax

- Pseudo-classes are always denoted by a single colon (:) followed by pseudo-class name;
- Selector: pseudo-class-name {  
CSS property  
}
- Here we use the :hover-pseudo-class and the :focus-pseudo-class

### Pseudo-classes used on links:

- For HTML links it is common to use the following pseudo-classes.
- \* :link - styles unvisited links
- \* :visited - styles visited links
- \* :hover - styles a link on mouse over.
- \* :active → styles on activated link.



## Program -

```

<!doctype html>
<html>
<head>
<title> Pseudo - class used on link </title>
<style>
a: link {
 color: blue;
}
a: visited {
 color: green;
}
a: hover {
 color: red;
}
a: active {
 color: pink;
}
</style>
</head>
<body>
<a href = "https://www.google.com"
target = "" - blank ""> Google
</body>
</html>

```

## Example on pseudo - class Focus

```

<!doctype html>
<html>
<head>
<style>
input: focus {
 background-color: pink;
}
</style>
</head>
<body>
<h2> using: focus </h2>
<p> use: focus to style an input field when
it gets focus: <input>
</p>
</body>
</html>

```

## CSS Navigation Bars

- CSS navigation bars are an important component of web design. Navigation bars helps users to easily navigate thru different section of a website.
- Navigation bars are typically built with HTML list elements (<ul> & <li>) and the styled with CSS to get a great look.
- Navigation bar are typically located at the top of the side of the webpage.





-Types

- ① Vertical navigation bar
- ② Horizontal navigation bar

### ① CSS Vertical Navigation Bar

- In a vertical navigation bar, the navigation links are stacked vertically (on top of each other) and is typically aligned along the left or right side of a webpage.
- The basics of a vertical navigation bar is an unordered list (`<ul>`) with list items (`<li>`) each holding a link (`<a>`)

```
<!doctype html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
ul {
```

```
margin: 0;
```

```
padding: 0;
```

```
width: 100px;
```

```
background-color: #ffffff;
```

```
}
```

```
o {
```

```
display: block;
```

```
color: black;
```

```
padding: 3px 10px;
```

```
}
```

1. Change the link and background color on the hover

```
a: hover {
```

```
background-color: black;
```

```
color: white;
```

```
</style>
```

```
</head>
```

```
<body>
```

```
 <div id="nav">
```

```

```

```

```

```
 Home
```

```

```

```
 News
```

```

```

```
 Contact
```

```
 </div> </body> </html>
```

```

```

```
 About
```

```

```

```
</div>
```

### ②

CSS Horizontal Navigation Bar

- \* In a horizontal navigation bar the navigation links are stacked horizontally (next to each other) and is typically aligned on the top of a webpage.

- \* The basics of a horizontal navigation bar is an unordered list (`<ul>`) with list items (`<li>`) each holding a link (`<a>`)

- \* It is also common to add a `<div>` element around the `<ul>` element that will serve as a container for your navigation bar.



## Horizontal NavBar using float

- \* One way of creating a horizontal navigation bar is to add the float property to <div> elements
- \* We can create a basic horizontal navigation bar with a dark background color and change the background color of the links when the user moves the mouse over them

Example:-

```
<doctype html>
<html>
<head>
<style>
body {
```

```
margin: 0;
padding: 0;
}
```

```
h1, p {
margin: 10px;
}
```

```
ul {
list-style-type: none;
margin: 0;
padding: 0;
}
```

```
overflow: hidden;
background-color: #233333;
```

```
li {
float: left;
}
```

a: hover { display: block; }

color: white;

text-align: center;

padding: 14px 16px;

text-decoration: none;

```

a: hover {
background-color: #231111;
}
```

```
</style>
</head>
```

```
<body>
```

```

```

```
<a href = "/pages/htm1" target = "" - blank
> Home
```

```
<a href = "/pages/htm1" target = "" - blank
> News
```

```
<a href = "hover.htm" target = "" - blank
> Contact
```

```
<a href = "margin.htm" target = "" - blank
> About
```

```

```

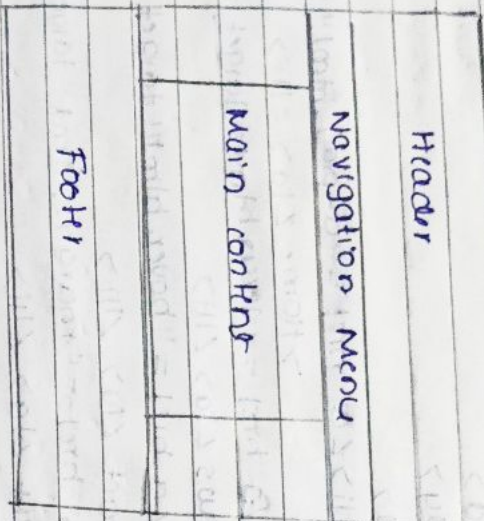
```
</body>
```

```
</html>
```



# CSS Website Layout

- CSS website layout divides a webpage into multiple sections like header, navigation bar, content area and footer, making it easier to organize content and build the site.
- There are tons of different layout design to choose from.
- However, the structure below is one of the most common



## Header Section

- The header is the top section of a webpage, typically containing the website's name or logo

## Navigation Menu:

is used to create a set of links for easy website navigation

## Content section:-

is used to add content to the page and you can adjust it according to the screen size

## Footer Section

The footer section is placed at the bottom of the webpage and typically contains information like contact, details, copyright or about us

Example:-

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
header {
```

```
background-color: #111111;
```

```
padding: 10px;
```

```
text-align: center;
```

```
}
```

```
footer {
```

```
padding: 5px;
```

```
background-color: #444444;
```

```
color: white;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
div to be wizard </div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```

```
</div>
```



```

</body>
</html>

```

#### Lab program - 4

#### ④ Designing of webpage - Document layout.

```

<html>

```

```

<head>

```

```

<style>

```

```

<body>

```

```

font-family: sans;
margin: 10px;

```

```

background-color: #f2f2f2;

```

```

</body>

```

#### ③ header &

```

background-color: #005999;
color: white;
text-align: center;
font-size: 24px;

```

#### ② nav a: hover &

```

background-color: #003366;
background-color: #003366;

```

```

</section>

```

```

background-color: white;
padding: 24px;

```

```

</body>

```

```

</html>

```

```

background-color: #0070C0;
color: white;

```

```

text-align: right;

```

```

</body>

```

```

</head>
</body>

```

```

</header>

```

```

My webpage layout
</header>

```

```

</nav>

```

```

 Home

```

```

 About

```

```

 Service

```

```

 Contact

```

```

</nav>

```

```

</section>

```

```

</body>

```

```

</section>

```

```

</body>

```

```

</html>

```

```

</body>

```

```

</html>

```

```

</body>

```

```

</body>

```

```

MyWebPage layout

```

```

Home About Service Contact

```

```

Welcome to my webpage.

```

```

This section contains the main content of the
webpage

```

```

Copyright: 2025 MyWebPage Mail: My123@gmail.com

```



# CSS POSITIONING

- CSS positioning is about controlling the placement of elements within a webpage.
- With CSS positioning, you can override the normal document flow.

## CSS POSITION PROPERTY

- The position property specifies the positioning type for an element.
- This property can have one of the following values.

- ① static → this is default
  - ② relative
  - ③ fixed
  - ④ absolute
  - ⑤ sticky
- elements are then positioned to their final location with the top, bottom, left & right property.

## ① CSS POSITION → static

- All HTML elements are positioned static by default.
- Static positioned elements are not affected by the top, bottom, left and right property.
- An element with position: static, is always positioned according to the normal flow of the page.

img of  
position: static;  
top: 50px  
left: 50px

## ② CSS POSITION — relative

- An element with position: relative, is positioned relative to its normal position in the document flow.
- Setting the top, right, bottom and left property will cause the element to be adjusted away from its normal position.
- Other content will not be adjusted to fit in any gap left by the element.

## ③ CSS POSITION — fixed

- An element with position: fixed, is positioned relative to the view port which means it stays in the same place even if the page is scrolled.
- The top, right, bottom and left property are used set the final location of the element.
- A fixed element does not leave a gap in the page where it would normally have been located.

## ④ CSS POSITION — absolute

- An element with position: absolute, is positioned relative to the nearest positioned ancestor (with position other than static).
- However, if an absolute positioned element has no positioned ancestors, it uses the document body, and moves along with page scrolling.



## 5) CSS POSITION - sticky

- An element with position: sticky, toggles between a relative and fixed position, depending on the scroll position.
- A sticky element is positioned relative until a certain scroll position is reached - then it "sticks" in that place (like position: fixed).

### Lab Program - 14.

14) Working with positioning and block properties

```
<html>
<head>
<style>
body {
background-color: lightblue;
p {
position: relative;
top: 50px;
left: 50px;
h1 {
position: fixed;
top: 50px;
div {
position: absolute;
top: 0px;
left: 0px;
img {
position: sticky;
top: 10px;

```

```
background-color: lightblue;
p {
position: relative;
top: 50px;
left: 50px;
h1 {
position: fixed;
top: 50px;
div {
position: absolute;
top: 0px;
left: 0px;
img {
position: sticky;
top: 10px;

```

</style>

</head> <body>

<h1> This is CSS POSITIONING </h1>

<p> This is paragraph consist of block property </p>

<div> This is div tag </div>

<img src = "rose.jpg" alt = "sample image" </img>

</body>  
</html>

</body> OR. body {background-color: lightblue;}

<html>

<head>

<style> body {background-color: lightblue;}

</style>  
</head> <body>  
<h1> This is CSS POSITIONING </h1>  
<p> This is paragraph consist of block property </p>  
<div> This is div tag </div>  
<img src = "rose.jpg" alt = "sample image" </img>  
</body>  
</html>



## CSS IMAGE SPRITES

✓ An image sprite is a collection of various small images put into one larger image file, called a "sprite image".

- A sprite images is typically arranged in a grid-like way.
- A web page with multiple images takes a longer time to load and generates multiple server requests.
- So, instead of downloading each image separately, the browser downloads the single sprite image file, which will reduce the number of server requests and reduce bandwidth usage.

### CSS ATTRIBUTE SELECTORS

- CSS attribute selectors are used to select and style HTML elements with a specific attribute or attribute value or both.

#### CSS [ATTRIBUTE] SELECTOR

- The [attribute] selector is used to select elements with a specific attribute.

<doctype html>

<html>

<head>

<style>

a[target]

background-color: yellow;

}

</style>

</head>

<body>

<h2> CSS [attribute] selector </h2>

✓> Select and style the links with a target attribute: < a >

<a href = "lab 14.html" target = " \_blank">

Java </a>

<a href = "display.html" target = " \_blank">

HTML </a>

</body>

</html>